MZUZU UNIVERSITY

FACULTY OF SCIENCE, TECHNOLOGY AND INNOVATION

DEPARTMENT OF MATHEMATICS AND STATISTICS

1.	PROGRAMME:	Bachelor of Science
2.	SUBJECT:	Mathematics
3.	LEVEL OF STUDY:	1
4.	COURSE TITLE:	Precalculus
5.	COURSE CODE:	MATH 1101
6.	DURATION:	16 Weeks
7.	PRESENTED TO:	Senate
8.	PRESENTED BY:	Dean of Education
9.	LECTURE HRS PER WEEK:	4
10.	TUTORIAL HRS PER WEEK:	1
11.	PRACTICAL HOURS:	0
12.	INDEPENDENT LEARNING HRS PER WEEK:	10
13.	TOTAL COURSE CREDITS :	12

14. PRE-REQUISITE COURSE CODES: None

15. CO-REQUISITE COURSE CODES: None

16. DELIVERY METHODS:

16.1 Mode of Delivery: Face- to Face and ODL

16.2 Teaching Methods: Lectures, tutorials, group discussion, projects, demonstration, brainstorming, problem solving

17. ASSESSMENT METHODS: Two continuous tests
One end of semester examination

18. ASSESSMENT WEIGHTING: 40% Continuous
60% End of semester

19. AIM(S) OF THE COURSE:

- To normalize the knowledge that students arriving possess in Mathematics
- Develop an intuition in and facility in using mathematical models

- Prepare students for subsequent courses in Calculus

20. LEARNING OUTCOMES:

A successful learner from this course will be able to:

- Use language of sets to conceptualize mathematical problems
- Solve equalities and inequalities
- Graph equations and functions
- Solve optimization problems without the use of Calculus
- Apply methods learnt in the context of logarithmic and exponential functions

21. TOPICS OF STUDY:

Review

- Sets, *n*th-roots, etc

Fundamentals

- Sets of real numbers
- Absolute value
- Rectangular coordinates
- Equation of a circle
- Symmetry
- Inequalities
- Quadratic inequalities
- Other types of equations

Functions

- The definition of a function
- The graph of a function
- Shapes of graphs – Average rate of change
- Techniques in graphing
- Methods of combining functions: Composition
- Inverse functions

Polynomial and rational functions

- Maximum and minimum problems
- Polynomial functions
- Rational functions

Exponential and logarithmic functions

- Exponential functions
- The exponential function $y = e^x$
- Logarithmic functions
- Properties of logarithms

- Equations involving logarithms and exponents

Trigonometry

- Algebra and the trigonometric functions
- Radian measure
- Radian measure and geometry
- Trigonometric functions of real numbers
- Graphs of sine and cosine functions including $y = A\sin(Bx+C)$ and $y = A\cos(Bx+C)$
- Trigonometric identities
- The addition formulars
- The Double-Angle formulas
- Trigonometric equations
- The inverse trigonometric functions

22. **Prescribed Texts**

- Cohen D et al, Precalculus: A problem oriented approach, Brookes/Cole Publishing, 2005

23. **Recommended Texts**

- E.W. Swokowski and J.A. Cole, *Precalculus: Functions and Graphs*, Brookes/Cole Publishing, 1998
- E.W. Swokowski and J.A. Cole, *Precalulus*, Thomson/Cole, 2008
- R. Larson, Precalculus: Functions and Graphs, Boston: Houghton Mifflin, 2005

This course outline was approved by Senate on 24th August, 2016